



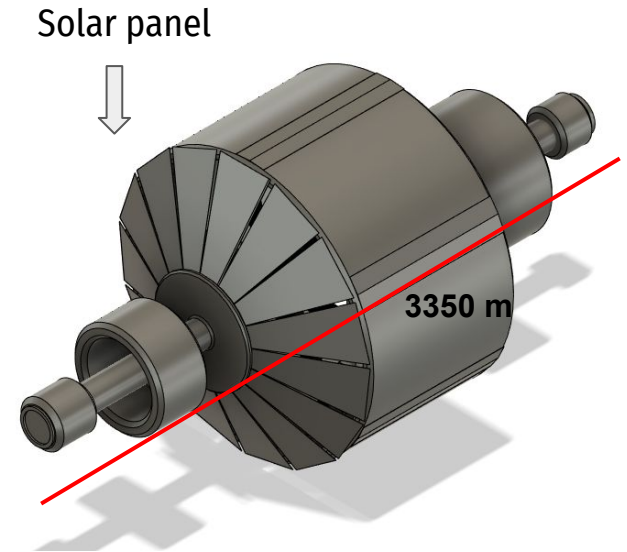
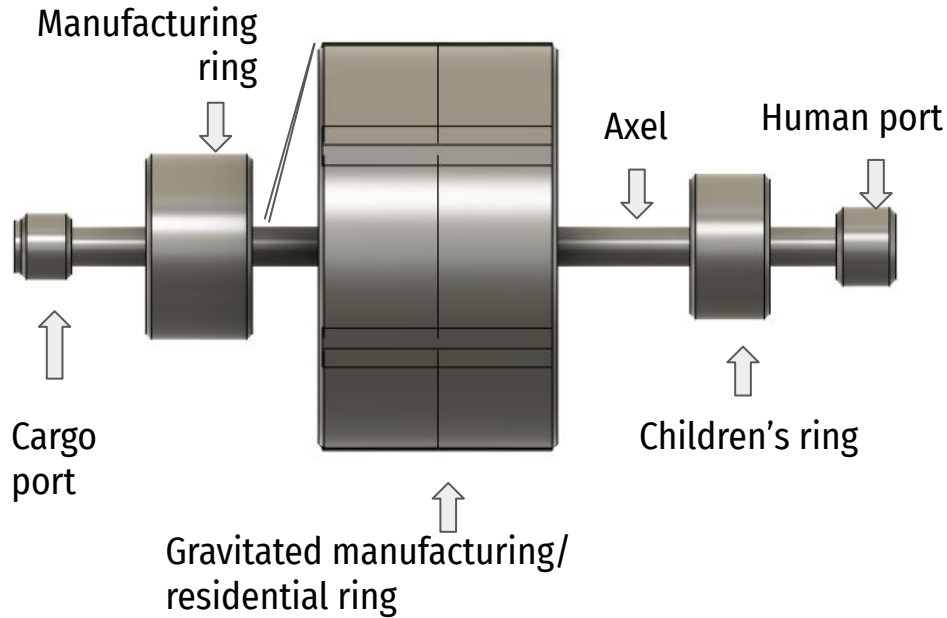
G

AYNAH

Grumbo Aerospace

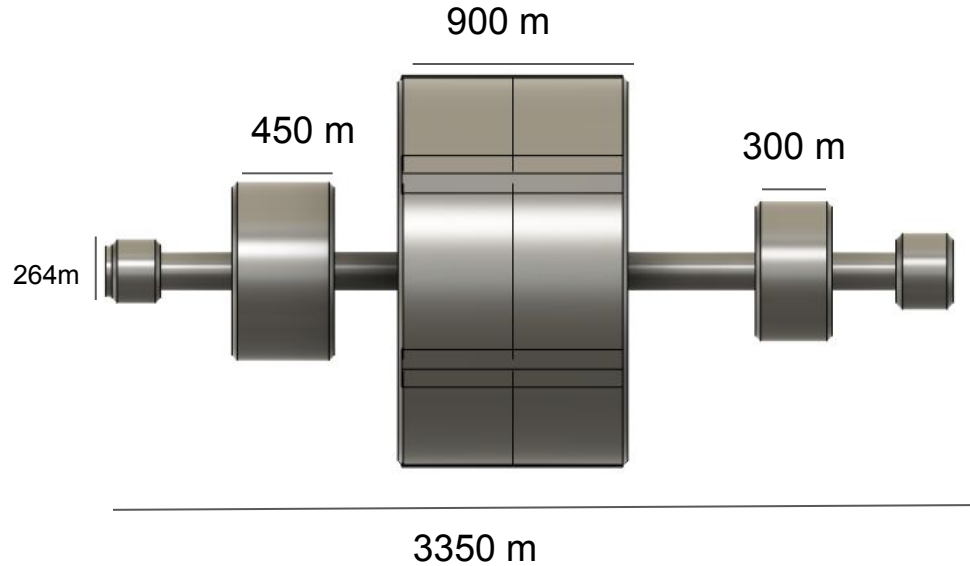


Aynah Design

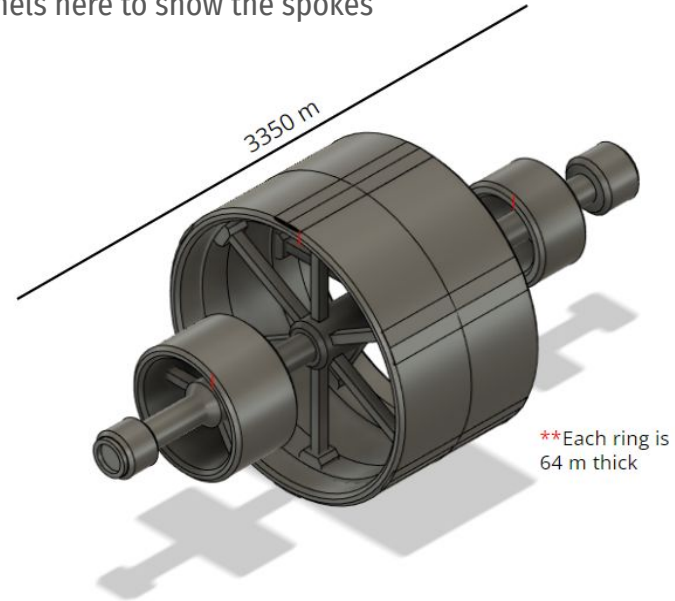


Models by Ryder Warbrick

Aynah Dimensions

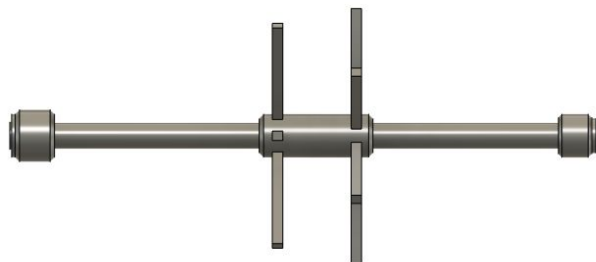


**Aynah is shown without the solar panels here to show the spokes

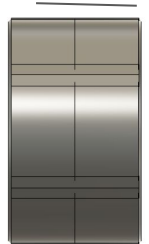


Models by Ryder Warbrick
Labeled by Amanda Nguyen

Dimensions and gravity calculations



900m



300m



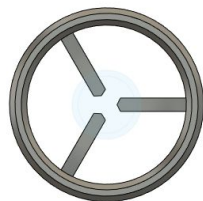
300m



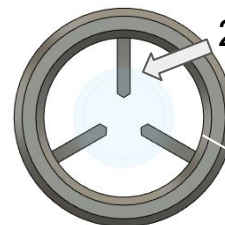
550m



1527m



550m



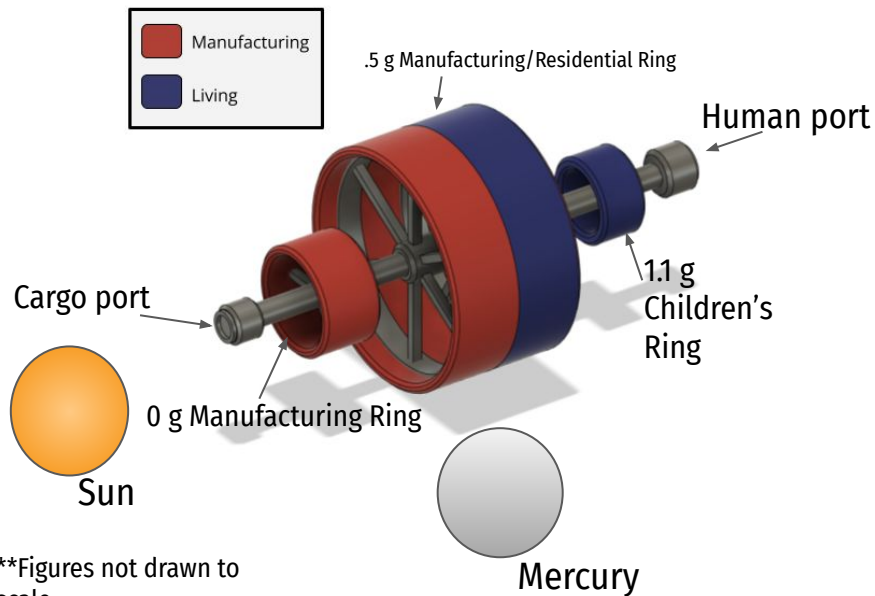
225m

64m

Large Ring:
0.765 rotations per minute
creating .5g.
 Smaller Ring:
1.674 rotations per minute
creating 1.1g.
 Manufacturing: 0g

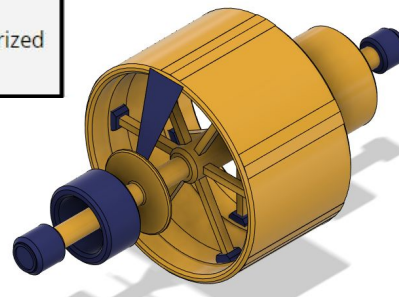
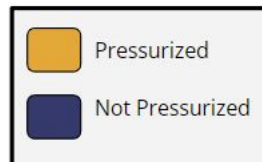
Distinguishing Areas on Aynah

Manufacturing vs. Living

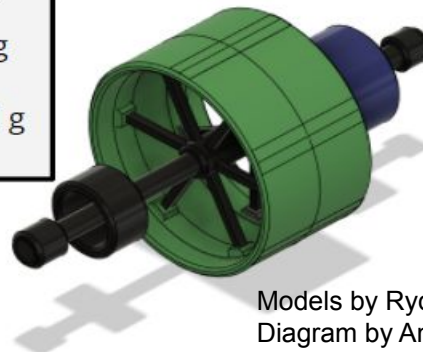
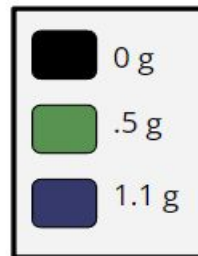


**Figures not drawn to scale

Pressurized vs. Unpressurized

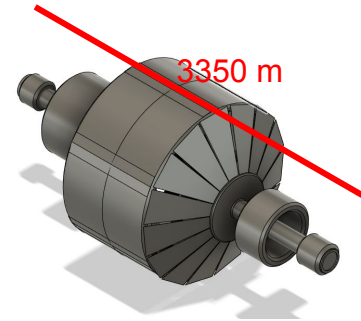


Different Levels of Gravity

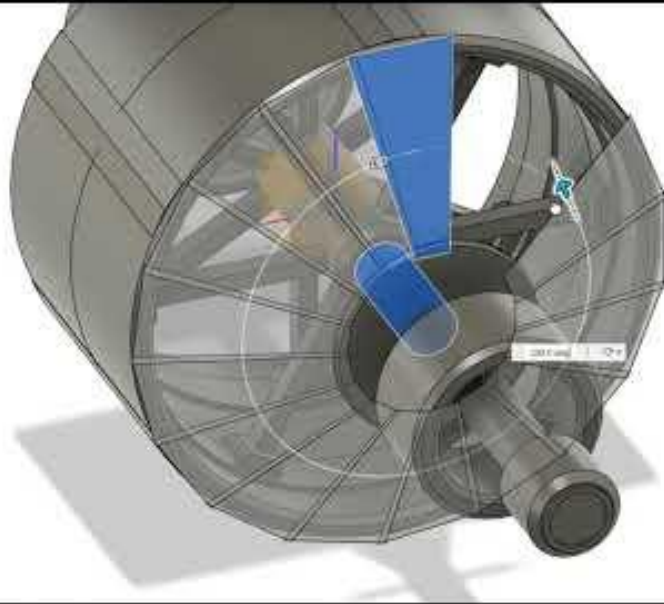


Models by Ryder Warbrick
Diagram by Amanda Nguyen

Solar Panel Shanzi

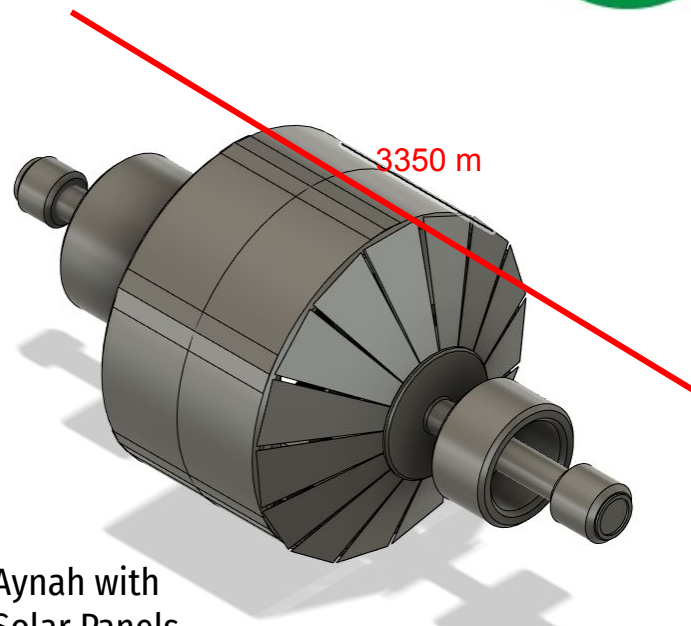
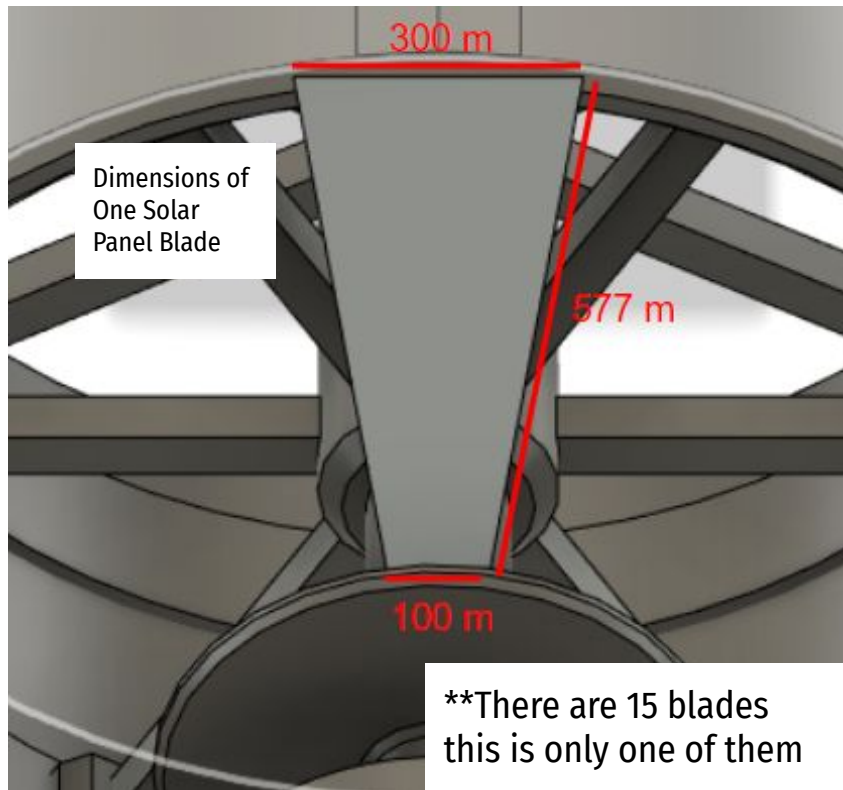


- The Solar Panel is designed like a chinese fan called “Shanzi” shown in the video.
- It is strategically placed in front of the residential areas where humans live to act as additional radiation shielding.



Models by Ryder Warbrick
Video Edited by Amanda Nguyen

Dimensions of Solar Panel Blades



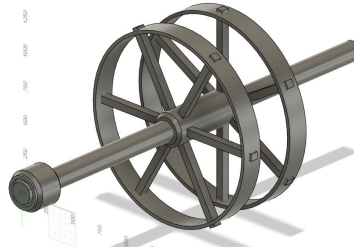
Aynah with Solar Panels

Models by Ryder Warbrick
Labeled by Amanda Nguyen

Construction Steps



Model by Ryder



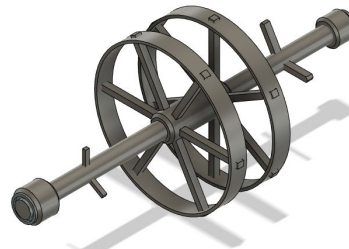
Model by Ryder

1st step:

- Build the main axle as well as the docking ports
- Programmed robots will be used

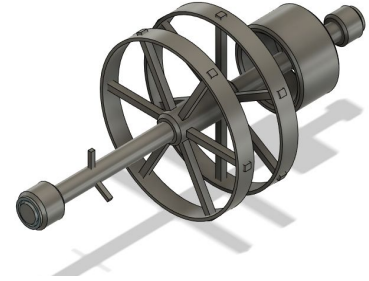
2nd step:

- Add support rings for the residential and manufacturing rings.
- Robots involved



3rd step:

- Spokes and rings of the children and zero G manufacturing ring were added
- Human port is added

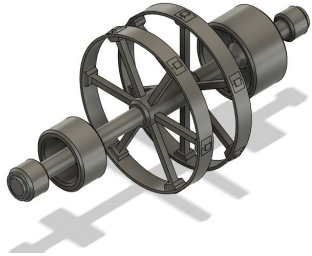


4th step:

- Children's ring is added
- Life support system is starting to function, ready for human to arrive

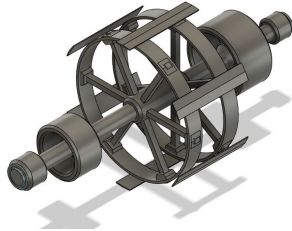
Models by Ryder Warbrick

More Construction Steps



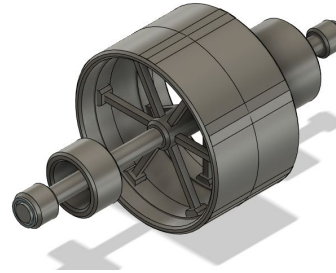
Step 5:

- The children's ring is capable for humans to monitor and direct the rest of the construction

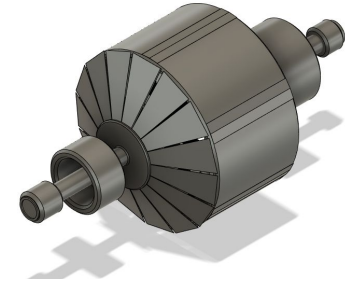


Step 6:

- More progress is made on the main ring - supporting ribs are added to give more stability



Step 7: add the residential/manufacturing ring

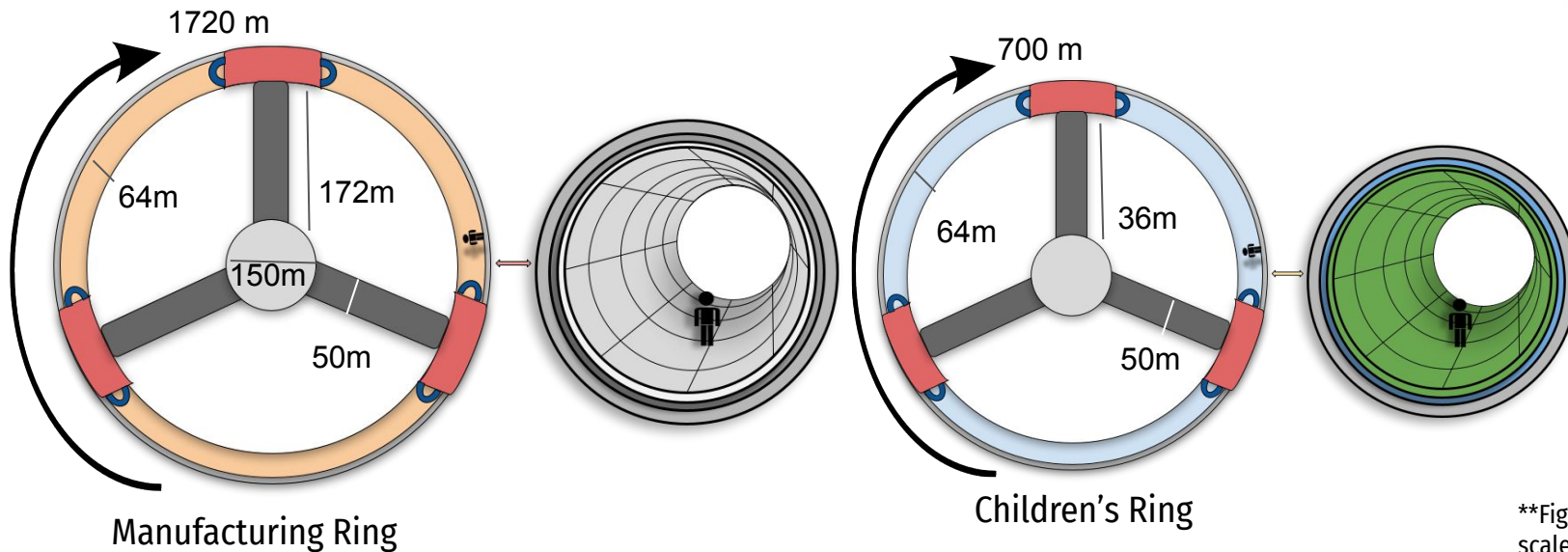


Step 8:

- SHANZI solar panels were installed.
- The progress is completed and the station is functional.

Models by Ryder Warbrick

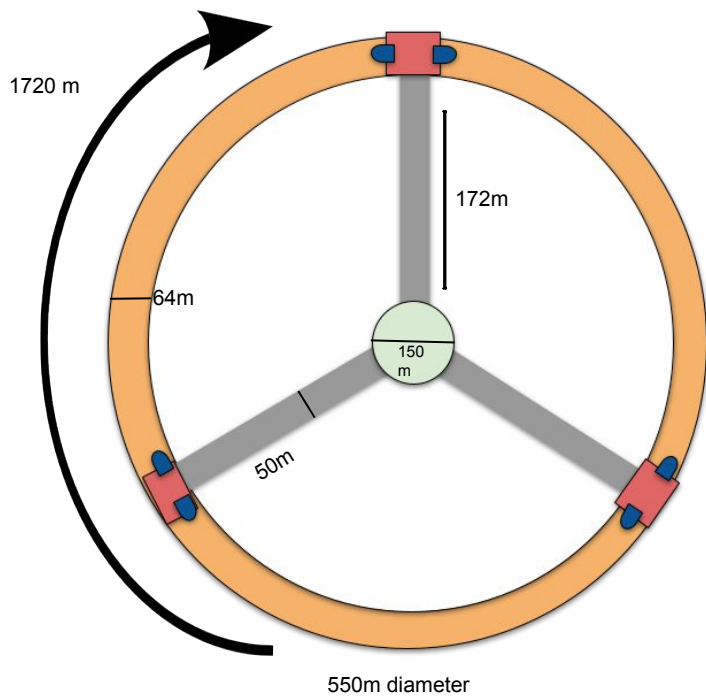
Down Surface Orientation



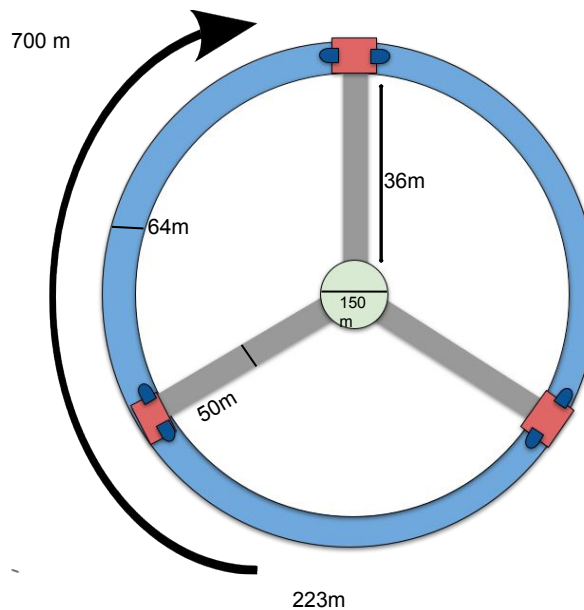
**Figures not drawn to scale

Diagram by Luke Perolino (Google Draw)

Down Surface Allocation



Manufacturing ring



Children's Ring

-  Axis
-  Spokes
-  Offloading Stations
-  Airlock Systems

Diagram by Luke Perolino, Google Draw

Down Surface of Split Residential/Manufacturing Ring

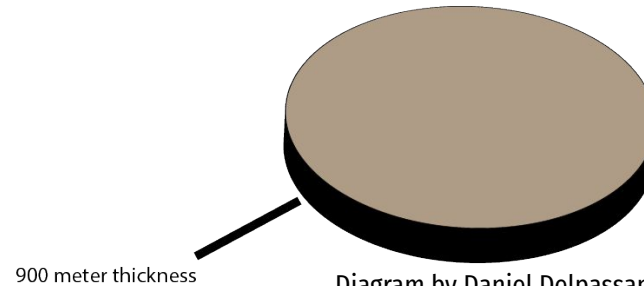
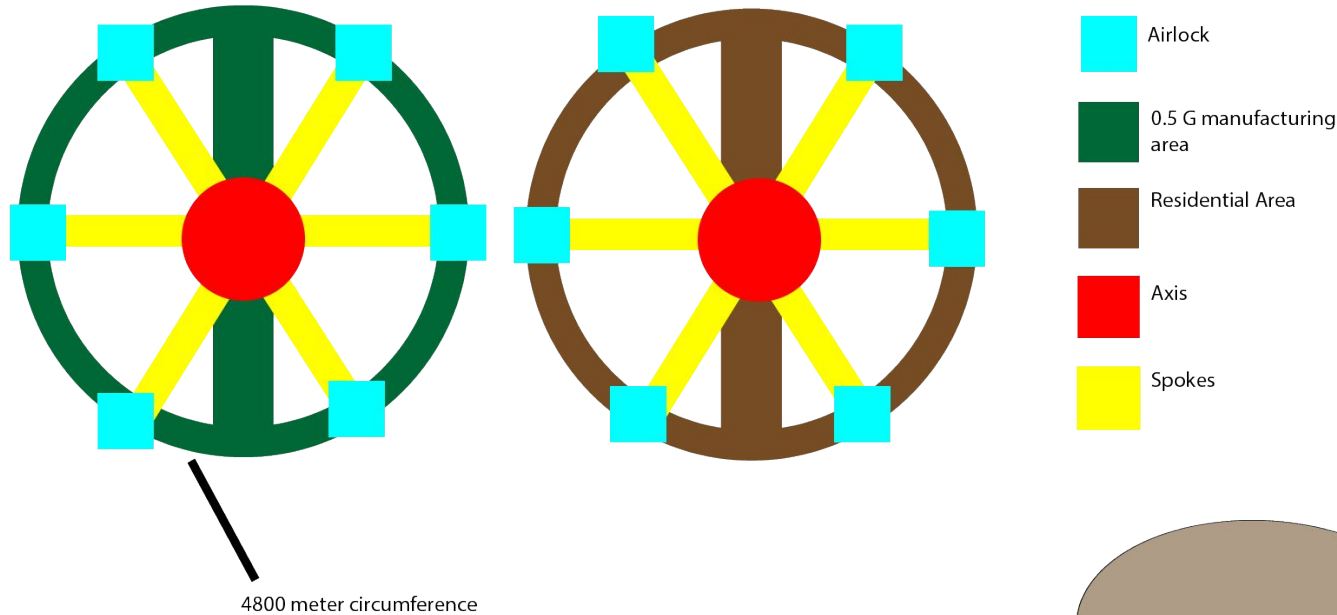
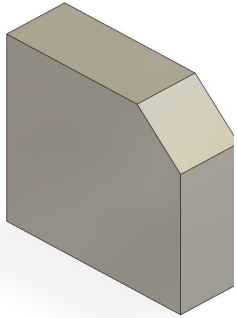


Diagram by Daniel Delpassand, Adobe Illustrator

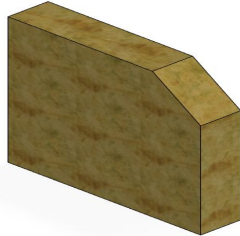
MAJOR STRUCTURAL MATERIALS LIST

MATERIAL SAMPLES

Titanium -
Strong, but
light.

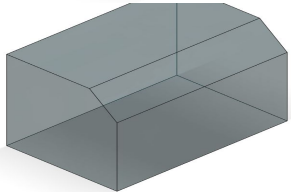


Pressure Glass -
Used to separate
independent living
spaces.

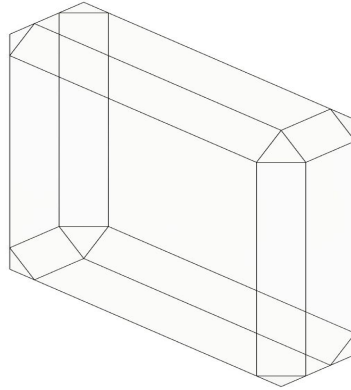
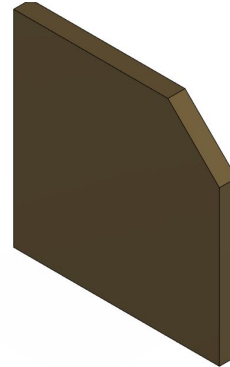


Buckystructure (above) to
protect against hull
breaches.

Water (left) - used to
protect against radiation



**Tantalum
Carbide** -
Protects against
extreme heat.



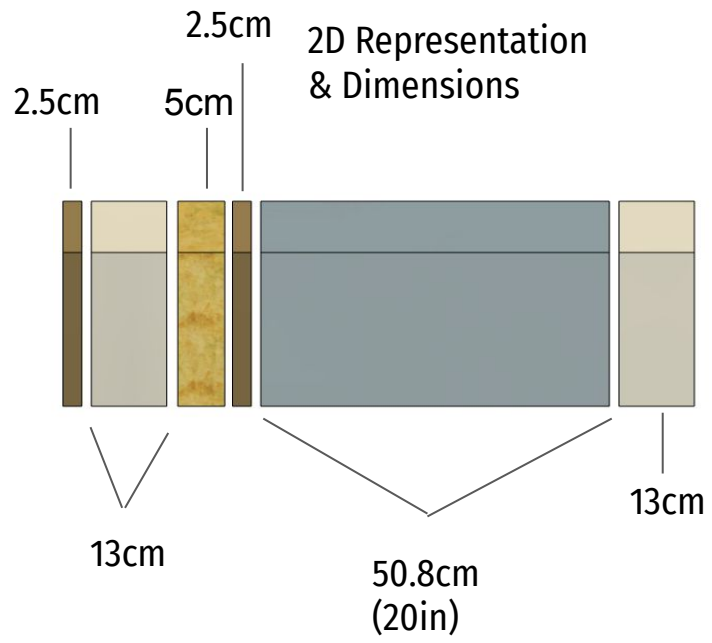
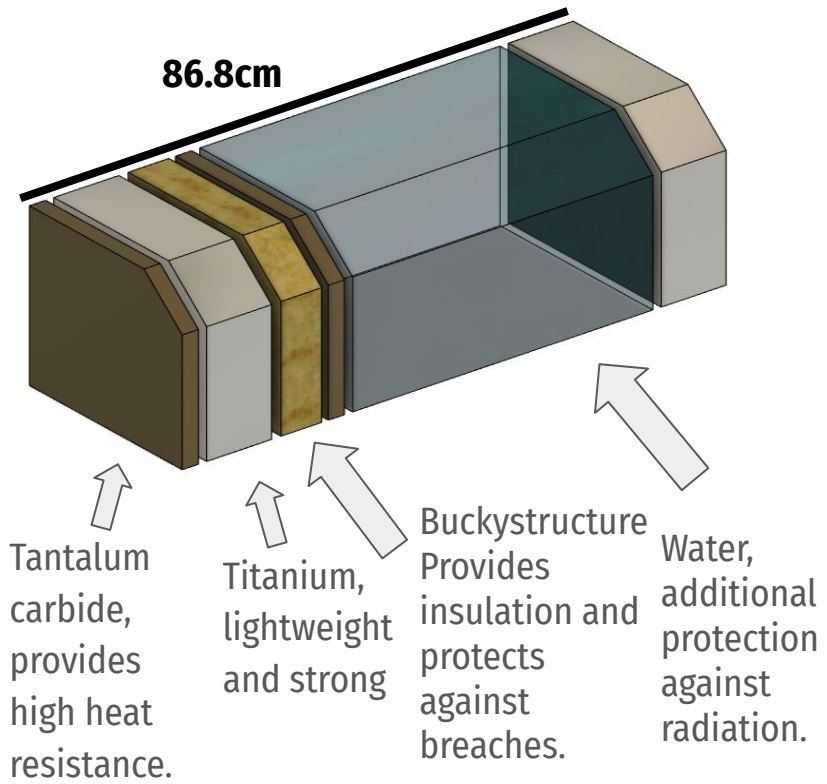
- The majority of materials will be sourced from Rockdonnell through the use of Litigation Limiters.
- Bellevistat in Earth-Moon L4 and BuckyBreakthroughs will supply other materials.



Tantalum Carbide	22000 m3	Earth
Pressure Glass	8,640 m3	Earth
Titanium	110,000 m3	Earth
Buckystructure	25,000 m3	Earth / BuckyBreakthroughs
Water	2.2B L	Earth

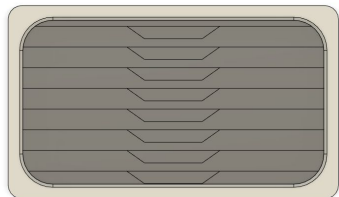
Models by Ryder Warbrick

RESIDENTIAL PROTECTION



Diagrams by Ryder Warbrick

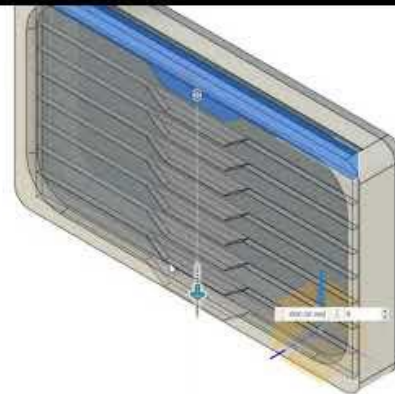
Natural View



Closed Version of
the water window.



Open version of
the water window



Water window demo

The window consists of 20 inches of water sandwiched between 2 panes of 2 inch ballistic glass. There is a fast closing shield that is comprised of 3 inch steel

Models by Ryder Warbrick
Video Edited by Amanda Nguyen

residential

hospitals/
clinic

commercial

offices

green space

recreational amenities/
entertainment

storehouses

food packaging
facility

control
rooms

spacesuit store

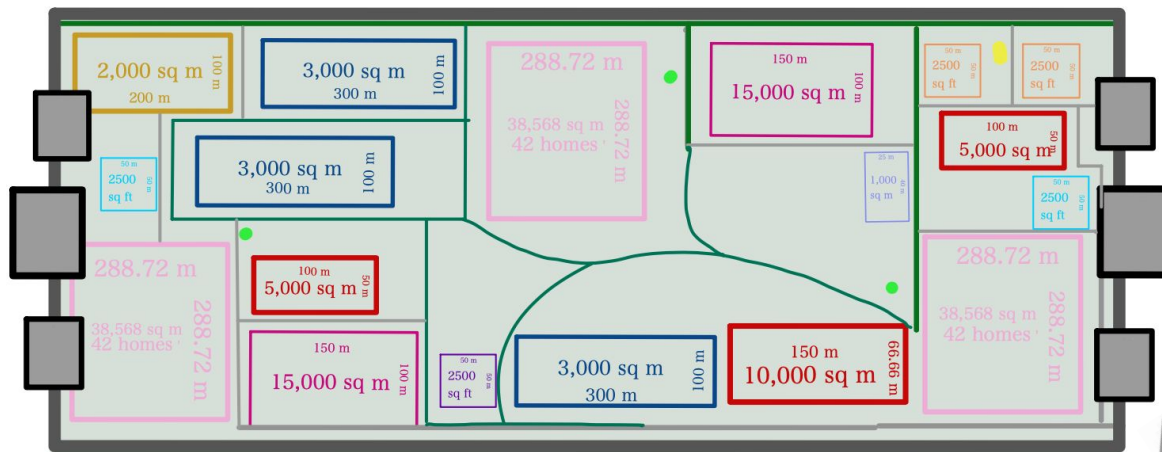
walking path

biking path

utilities

airlock

1600 m



Community map by Jana Salam

Total facilities add up to 190,204 sq m

Remaining 529,796 square meters
allows for spaciousness and long
lines of sight.



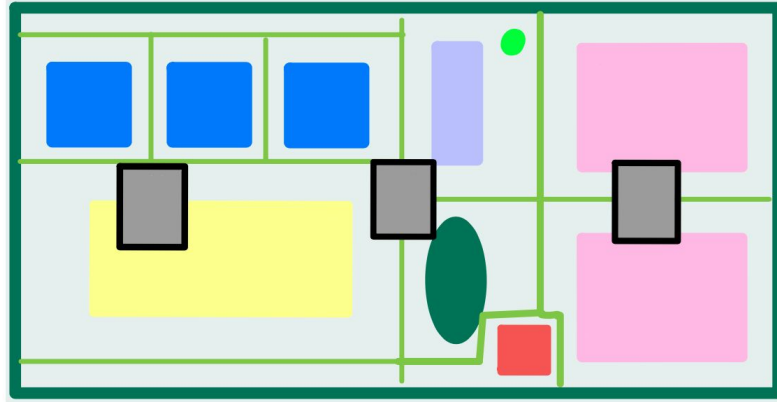
450 m

Goes to spokes

Goes through
different
sections

Children Community Design

700m



spacesuit
store

control
rooms

museum

nursery

entertainment

green space

schools

park

walking path

airlock

Community map by Jana Salam



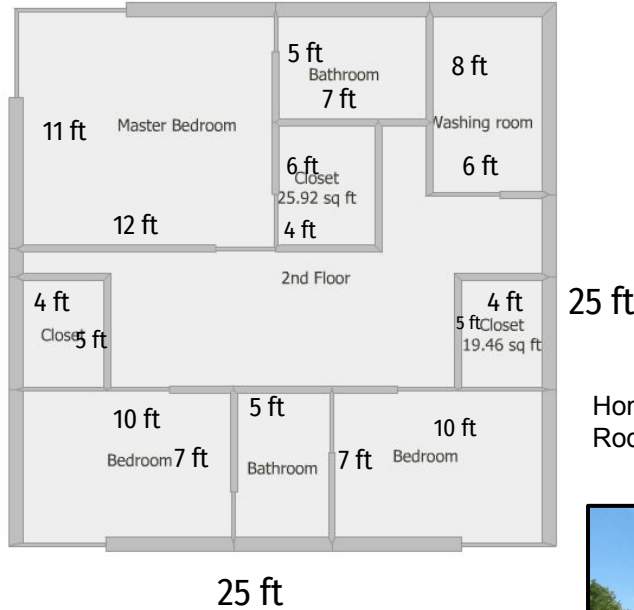
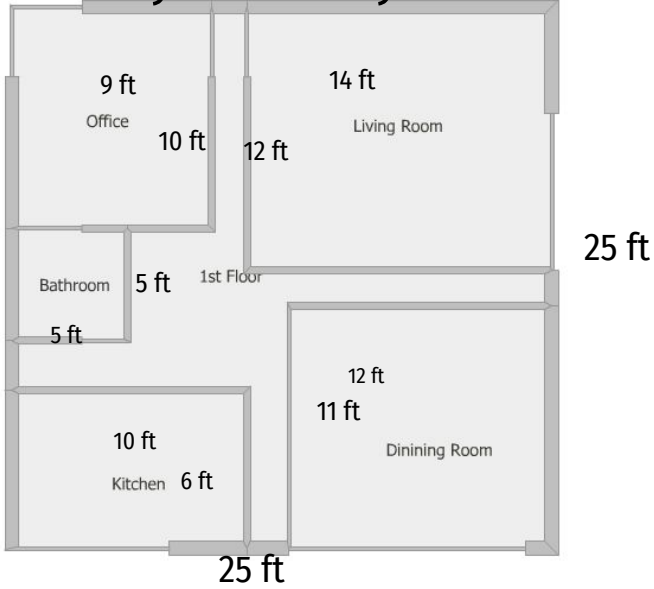
300m



Indoor Ski Area

Art by Jana Salam

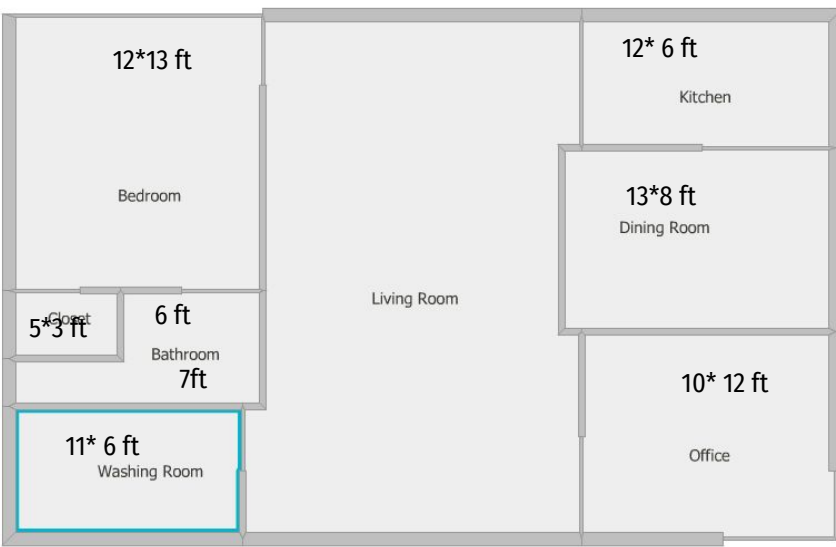
Family Home Layout



Home Designs by Sophie M., Esteffi Rodriguez, and Jenna Donahue



This is a two story layout meant for families with 2+ children. The thin lines are for windows and doors
 25x25 (2 floors)
 1250 sq.ft



25 ft

40 ft

25*40
1000 square feet
It is a one level flat.
This model is for a single adult.

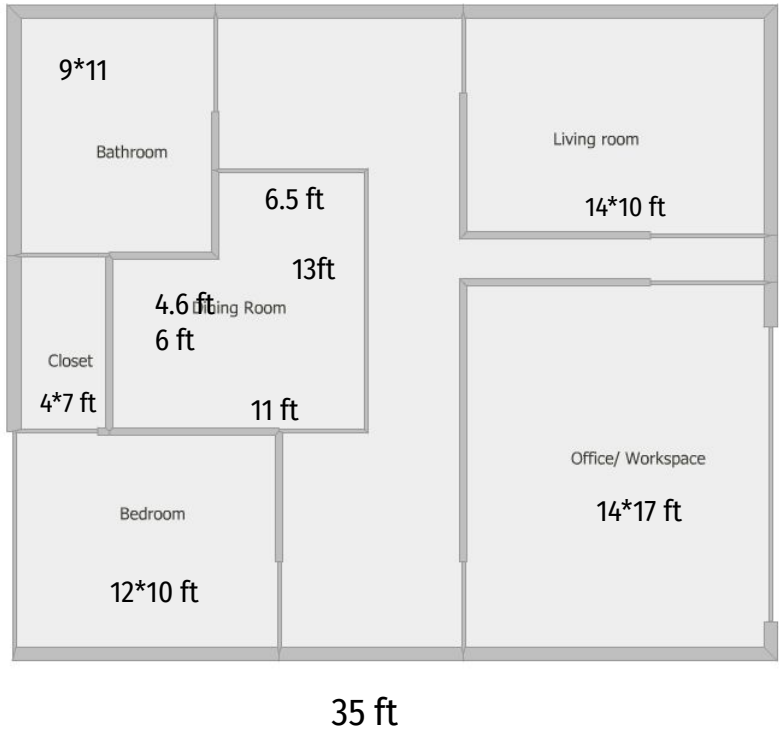


*Number of each	Type of floor plan
4900	Single Adult Homes
700	Family Homes
3300	Married Couple Homes
400	Temporary Living

Home Designs by Sophie Maxwell, Esteffi Rodriguez, and Jenna Donahue



This housing flat is only meant for temporary researchers residences. It comes to a total of *1050 square feet* and has one floor. *30*35 ft*

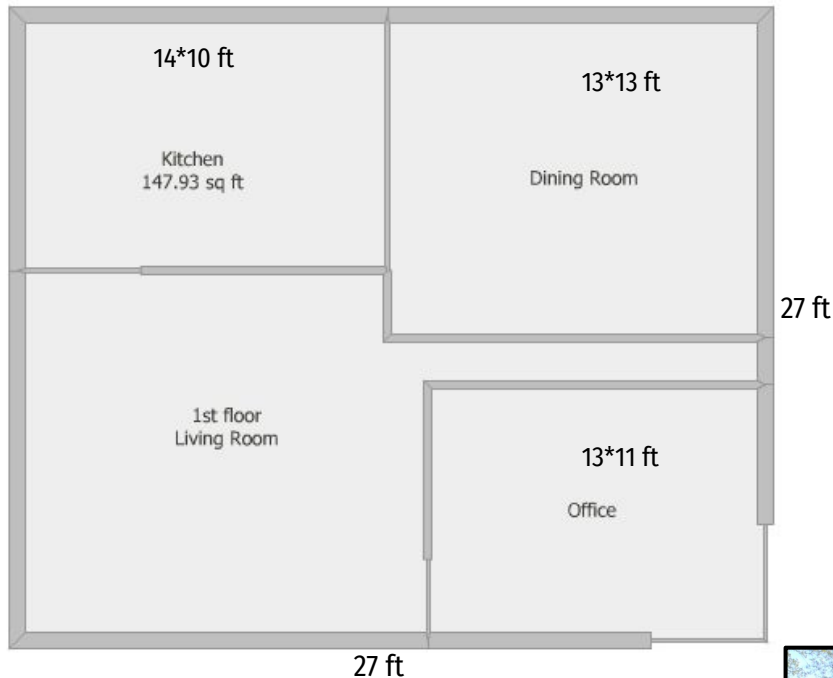
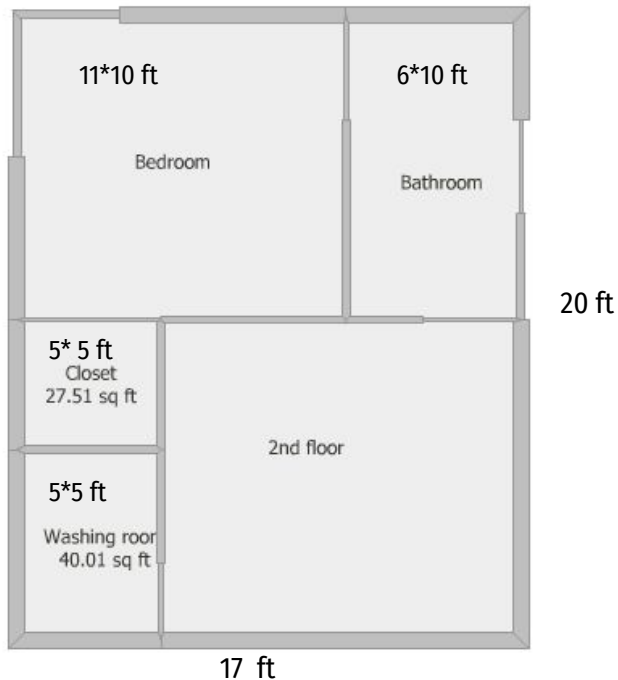


30 ft



There are extra of this type of residence for residents from other sections of the community to stay in during emergencies.

Home Designs by Sophie Maxwell, Esteffi Rodriguez, and Jenna Donahue



$(20*25)\text{ft} + (30*30)\text{ft} = 1400 \text{ Square feet}$. This is built mainly for couples that live together

Home Designs by Sophie Maxwell, Esteffi Rodriguez, and Jenna Donahue



Spacesuits



	Adult units	Child units	Description
Standard Suits	15,322 (max adults + 12%)	792 (total children + 10%)	Used during loss of pressure or during other emergencies.

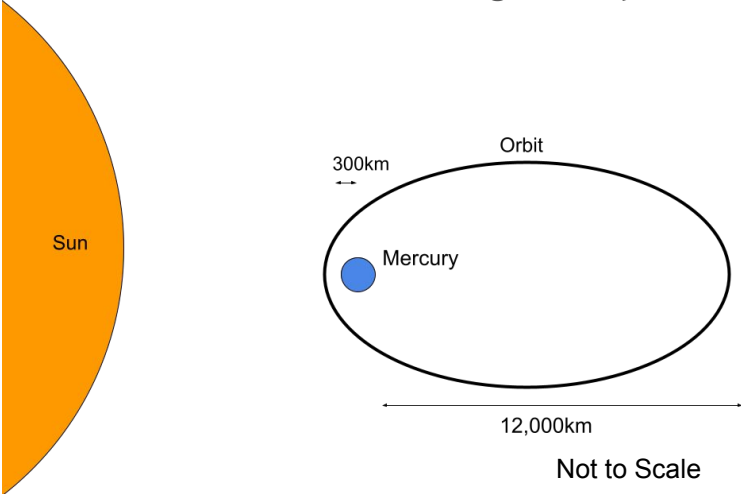
Spacesuit Table by Jana Salam, Google Docs

Subcontractor: Extreme Survival Technologies

- Spacesuits are stored in storage areas in front of each airlock
- Storage areas also serve as donning/doffing areas to ensure that people can easily access spacesuits as they leave their section settlement

Orbital Location

- Based on MESSENGER probe and ISS orbit
- Distances have been increased to reduce the number of required satellites for triangulation at further distances
- Minimizes the effect of gravity wells on larger ships docking at the settlement while also allowing for shorter distances when accessing Mercury



Perihermion Distance	300 km
Apohermion Distance	12,000km

Table by Jack Guo, Google Docs

Diagram by Jack Guo, Google Drawings

Food Production and Distribution



Types of Food

- Proteins
 - Chicken
 - Rabbit
 - Soy-based
 - Beans
- Grains
 - Millets
 - Kamut
- Dairy
 - Soy
- Fruits
 - Bananas
 - Melons
 - Berries
- Vegetables
 - Sweet Potatoes
 - Spinach
 - Green Onion
- And More!

Production

- Hydroponic system
- Plants ordered by those most to least drought tolerant
- Harvesting done by SCUDDER

Processing

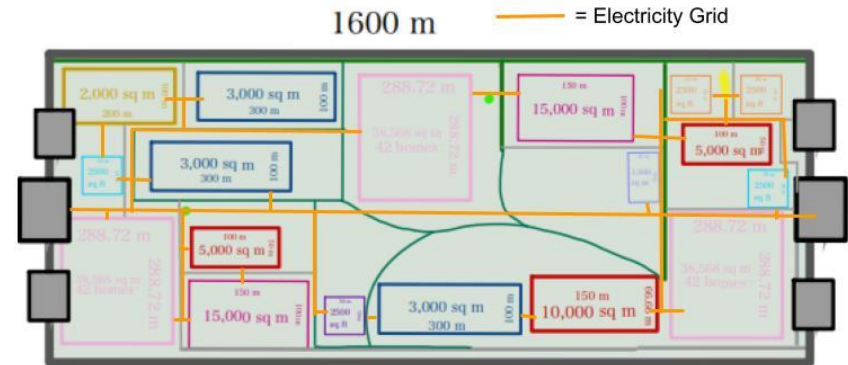
- After production, food will be brought to food warehouses where they would be processed, stored and sold in the residential areas

People/Time	Kilograms eaten
Average adult/Day	1.81kg
Average adult/Year	662.2 kg
14,400 adults/Year	9,536,326 kg

Subcontractors: Garden-A-Go-Go

Electrical Power Generation and Distribution

- 7485.12 Mwh per day
- 16414739 panels purchased initially from ZAP! Industries
- Energy includes residential, business, and 15% redundancy factor
- Excess energy can be stored in graphite-based batteries that can store up to a total capacity of 2 Mwh
- Batteries will also be stored in the warehouse



Community map by Jana Salam



Networking

Internal Communication

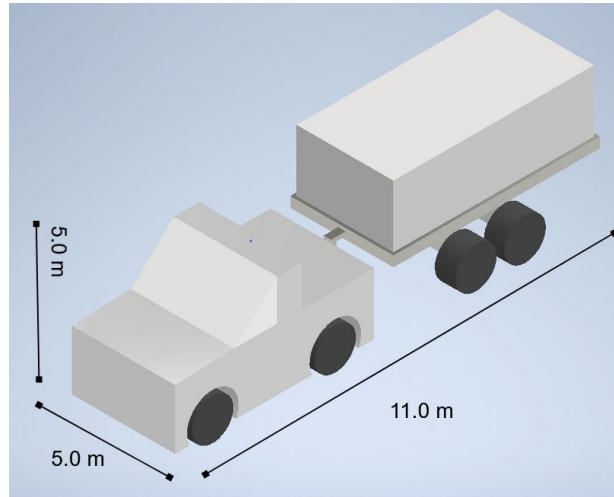
- Orbitlink communications fiber optics

External Communication

- Infrared Laser Communication - decreases communication time gap between Mercury and Earth
- Disruption-tolerant Networking - satellites hold onto information packets until connection is reestablished
- Satellite Relay System - Allow for satellites to communicate with ground stations without direct line of sight
 - Subcontract Dougeldyne through Litigation Limiters
- Proxy-cache system - Satellites will cache used websites for quicker access and monitor them for security

Internal Transport Systems

- Electric cars will be used to transport CASSCS and people around the city to desired locations.
- Elevator systems will provide transportation inside spokes of the settlement



Model by Donovan Munoz, Inventor



Atmosphere

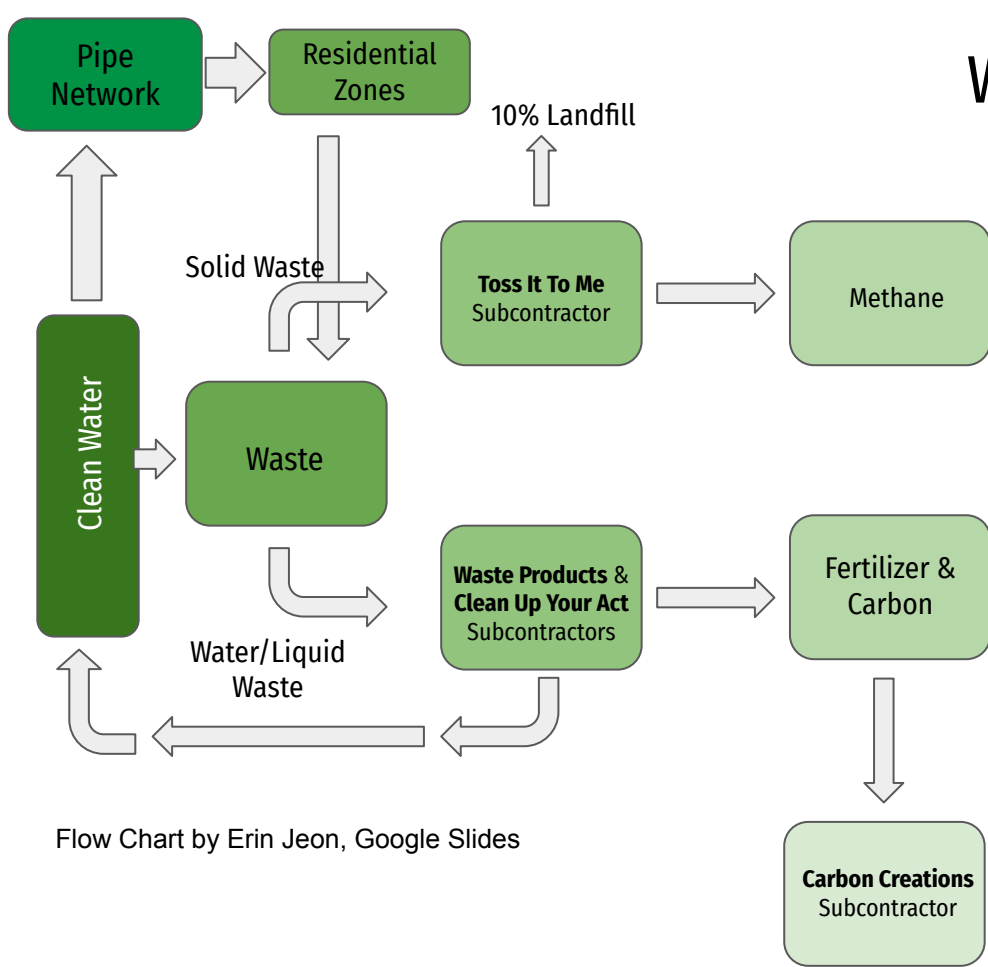
Elements	Total amount in kg	Average amount of g/L at .65 air pressure at sea level	Percentage	Used for	CASSC Numbers
Nitrogen	259,700,000	.53 g/L	67.855%	Filler	1,684,893
Oxygen	122,500,000	.25g/L	32.05%	Human Consumption	151,909
Carbon dioxide	166,600	.00034g/L	.044%	Temperature control and plants	17,028,721
Water Vapor	196,000	.00040g/L	.051%	Temperature control	3,747(Not included in water calculations)

Subcontractors: Clean Up Your Act & Stuff of Life

Table by Ariel Harrington, Google Docs



Waste and Water



Flow Chart by Erin Jeon, Google Slides

Drinking Water	2 L per person per day
Household Water Usage	300 L per person per day
Contingency (14 days)	60,883,200 L
Water Shield	2B L
Total CASSSCs of Water for Initial Shipment	11,854

Table by Erin Jeon, Google Slides

- 1% loss of water during recycling
- Sewer routing is done by various collaborating subcontractors: Waste Products and Clean Up Your Act



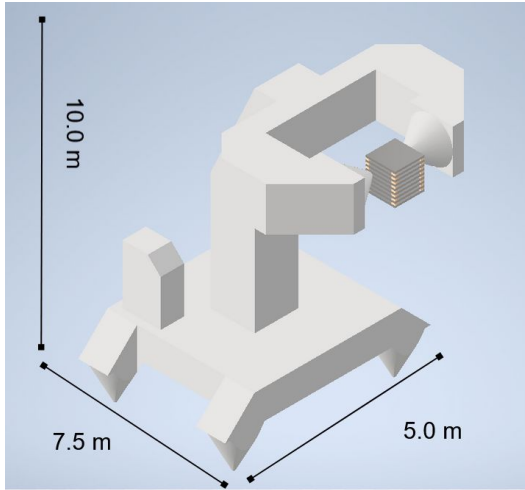
Automation in Construction

Name	Function	Quantity	Dimensions
Taggart	Exterior Construction	75	5.0 m x 7.5 m x 10.0 m
Galt	Interior Construction	75	3.5 m x 4.5 m x 6.0 m
Rearden	Installation of utilities and interior finishing touches	50	2.0 m x 2.0 m x 1.5 m
Wyatt	Communicates with control room, monitors construction automations	20	2.5 m x 2.5 m x 1.5 m
Danneskjold	Transports materials to construction sites	25	5.0 m x 5.0 m x 11.0 m
Conway	Jigs; platforms for automations to work on	45	6.0 m x 8.0 m x 1.0 m

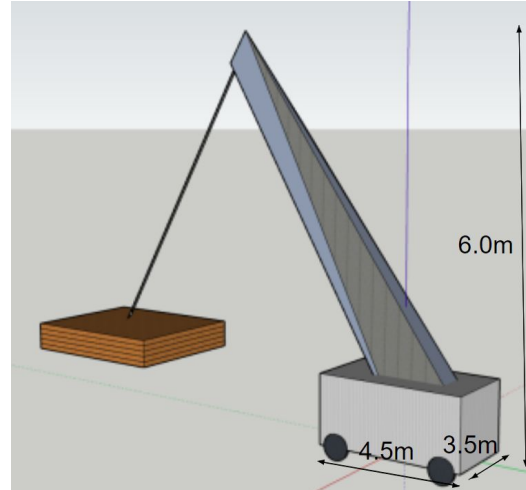
- All construction automations are mobile.
- Exterior and interior construction automations, as well as jigs, are modular and multifunctional
- Autonomous robots allow for safer and more efficient construction

Construction Automations Table by Ethan Lai, Google Docs

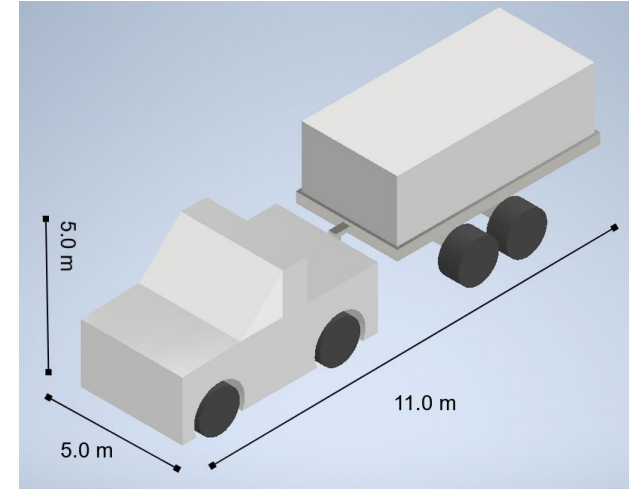
Construction Automation



Taggart Automation carrying steel plates
by Donovan Munoz, Inventor



Galt automation moving a plate by Jack
Guo, Sketchup



Danneskjold unit transporting a CASSSC by
Donovan Munoz, Inventor

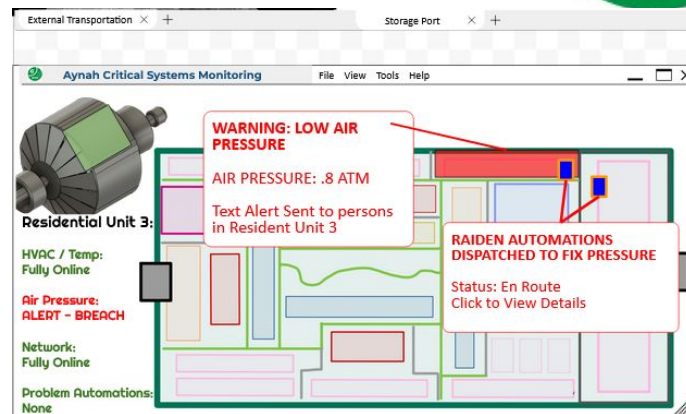
Control Rooms

During construction

- Crew members interact with construction automation from Earth in beginning stages
- Control room in the Children's Ring will be populated with crew members after stage 5. Crew will receive reports from **Wyatt** automations, who supervise other construction automations. In case of emergency, crew will be able to respond and intervene after being notified by **Wyatt** units.

After Construction

- 4 control rooms total: 1 room located in each segment of the main manufacturing/residential ring and 1 in children's ring
- Monitor their respective domains, keeping track of critical functions like pressure, power, and communication lines, as well as internal transportation of CASSCs and any hazard alerts.
- One of the 3 control rooms in main ring will also monitor 0g manufacturing ring, ensuring smooth production lines and proper transfer of CASSCs from cargo port to warehouses and back.

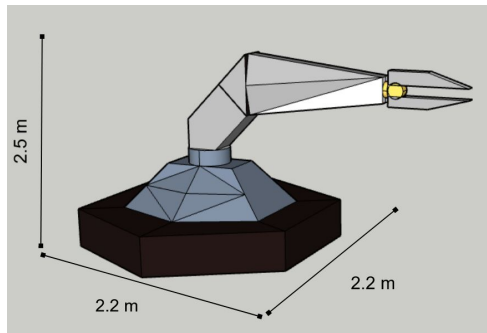


Sample Control Room Console by Minh Trinh

Automations for Business and Production

Name	Function	Quantity	Dimensions
Scudder	Maintain Garden-a-go-go tasks and duties	40	1.5 m x 2.0 m x 2.5 m
Narragansett	Food packaging	90	2.2 m x 2.2 m x 2.5 m
Denagger	Delivery of food items to residents	15,000	0.3 m x 0.3 m x 1.0 m

Business and Manufacturing Automations Table by Ethan Lai, Google Docs



Narragansett Automation by Ethan Lai, Sketchup

- Aid in food production, packaging, and delivery
- Reduce need for manual labor
 - **Scudder** units replace humans originally required to maintain Garden-a-go-go units



Automation for Maintenance and Repair

Name	Function
Taggart	Exterior Maintenance*
Galt	Interior Maintenance*
Rearden	Maintenance of utilities and interior finishing touches*
Wyatt	Communicates with control room, monitors maintenance automations*
Danneskjold	Transportation of materials*
Conway	Jigs*

Automations for Maintenance and Repair Table by Ethan Lai, Google Docs

*Automation are repurposed, so quantities and sizes are the same as 5.1

- Construction automations will be repurposed for maintenance and repair functions, due to their modular design.
- Reduces the need for production of more automations, decreasing cost



Alert Types and Safety Functions

Hazard	Detection Method	Contingency Plan
Automation Malfunction	Wyatt units detect abnormality in automation behavior	Wyatt reports back to control room, malfunctioned automation transported to control center to be repaired or repurposed
Loss of electricity	Energy management system	Backup generators activated; Rearden automations repair main generator
Loss of communications	Peer-to-peer communication; confirmation of messages sent and received	Redundant satellite relay system
Gas Hazard	Air samplers test composition of air for hazards	Air filters (Thompson) purify air; Rearden automation repair gas utility
Fire	Visual Imaging and infrared Fire Detector	Residents evacuated, fire suppression drones (Halley) deployed
Hull Breach and Depressurization	Pressure sensors	Rearden automations with pressure sensor attachment detects and fixes breach. Areas with critically low or high pressure are shut off.

Alerts and Safety Features Table by Ethan Lai, Google Docs

Hazard Level 1 (Green) – Minor inconvenience to settlement function, crew members notified

Hazard Level 2 (Yellow) – Hindrance to settlement function, crew member notified, automations deployed

Hazard Level 3 (Red) – Major threat to settlement, crew members and residents notified, automations deployed immediately



Security Measures

Security Level	Alpha	Beta	Gamma	Delta
Accessible Locations	Critical Settlement Systems (airlocks, power systems, etc.)	Control Rooms, settlement data, communication systems, manufacturing areas	Residential Units, Personal Devices, office spaces	Community Areas
Access Granted to:	Executive officers, crew supervisors	Settlement Officers, Crew members	Full-Time Residents	All residents and visitors
Security Measure(s)	Facial recognition, Finger-vein scan	Retinal Scan, Fingerprint Scan	Fingerprint scan	Key Card

Security Measure and Authorized Table by Ethan Lai, Google Docs

- 4 Levels of Security
- Biometrics of increasing complexity to ensure that the most critical locations are most secure
- Use of multi-factor authentication in levels Alpha and Beta to further enforce security



Everyday Automations

In the Household:

Name	Function	Dimensions	Quantity
Starnes	Cleans houses and open spaces, vacuuming and disinfecting	.45m x .45m x .2m	1 per household & office area
Mouch	Takes away clothes for laundry, cleans, folds, and delivers them	.5m x .5m x .5m	1 per household
Mulligan	Moves & reorganizes furniture	.5m x .5m x .5m	1 per household
Willers	Washes, dries, and stores any dishes	.6m x .5m x 1.0m (adjustable)	1 per household
Nealy	Customizable heating, ventilation and cooling	.3m x .3m x 1m	1 per household

- Subcontracted by Bots4U
- Stored in each residential and office unit

Household Automations Table by Minh Trinh, Google Docs



Everyday Automations (cont.)

Name	Function	Dimensions	Quantity
Denagger	Delivery & movement of personal goods on request of the owner	1.0 m x 1.0 m x .75m	15,000
Kellogg	Performs medical checkups, first-responder to emergencies until a human professional takes over, carries medical equipment	1.0 m x 2.0 m x .75m	70
Halley	Fire suppressing drone	.5m x .5m x 4.0 m	50
Thompson	Air Purification & Filtering	.1m x .2m x .4m	5,000

Community Automations Table by Minh Trinh, Google Docs

Name	Dimensions	Quantity
Wristwatch	10cm x 10m x 5cm	1800 (1 per person)
Mobile Communication Device	16 cm x 9cm x 1cm	1800 (1 per person)
Personal Laptop	20cm x 35cm x 2cm	1400 (1 per resident)
VR Set	10cm x 18cm x 10cm	1400 (1 per resident)

Personal Devices Table by Minh Trinh, Google Docs



Networks and Security

Decentralized network

- Unused computing resources of devices & automations will be meshed and contributed to the network, where any device will have unlimited access.
- Critical devices/automations for life support or data storage are redundant
- Persons using devices will have to provide appropriate security credentials
- Better data reconciliation as all the data will always be available to everyone at all times under encryption.

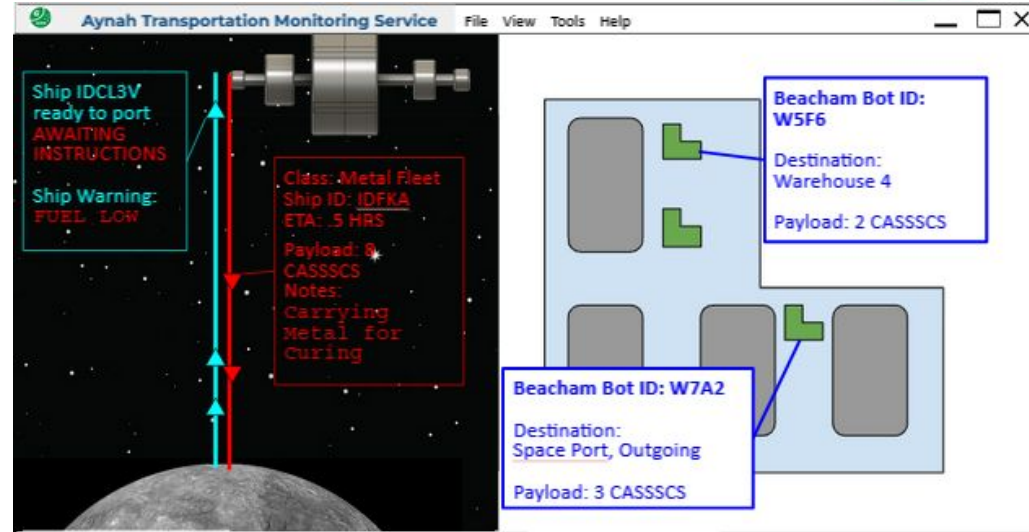
Security

- The use of a decentralized network protects against widespread network malfunction or practical cyberattacks
- Firewalls and malware-checking Intrusion detection systems will be implemented on all devices
- Transport layer security with ECDH, 1024-bit ECC will be implemented, secure against quantum attacks
- Data will be encrypted and stored redundantly, and regularly backed up
- To be GDPR-compliant, personally-identifiable-information is collected only by consent, and the rest is anonymised.

Automations in Reardonium Operations

Name	Function	Quantity	Dimension
d'Anconia	Loading and Unloading of CASSCSs	20	4.5 m x 4.5 m x 6.0 m
Beacham	Transportation of ore from cargo port to manufacturing ring	5	5.0 m x 5.0 m x 10.0 m
McNamara	Processing and quality control of reardonium parts	40	4.5 m x 4.5 m x 6.0 m

Reardonium Automations Table by Ethan Lai, Google Docs

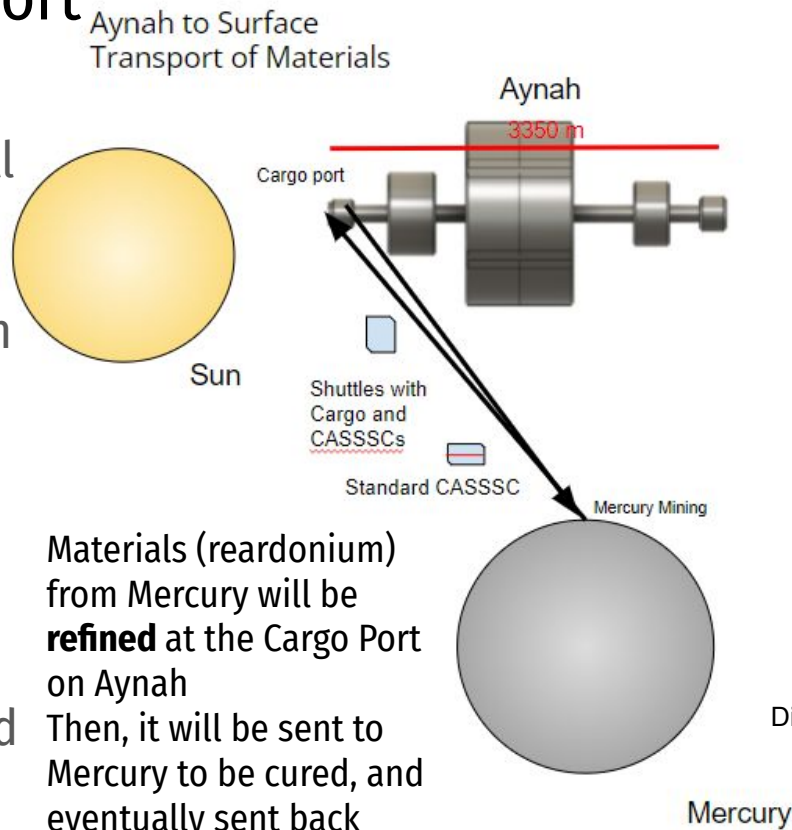


Sample Reardonium Control Room System by Minh Trinh

- Conveyor belt system to process and create reardonium parts
- Shuttles and transportation automation that carry reardonium ore and parts all have unique IDs that allow control center to track shipments with interface (shown above)

Aynah to Surface Transport

- Aynah's Cargo Port is set **facing towards the sun** at all times in the **polar orbit around Mercury**
- Between Mercury and Aynah will be the transport of materials in CASSSCs from Mercury's surface, including **reardonium** transport
- A fleet of landing shuttles launched from the ports and landing at the ports



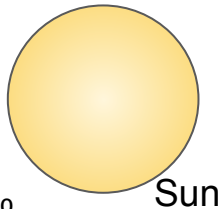
**Figures not drawn to scale

Subcontractor: Custom Cargo Accommodations for CASSSCs

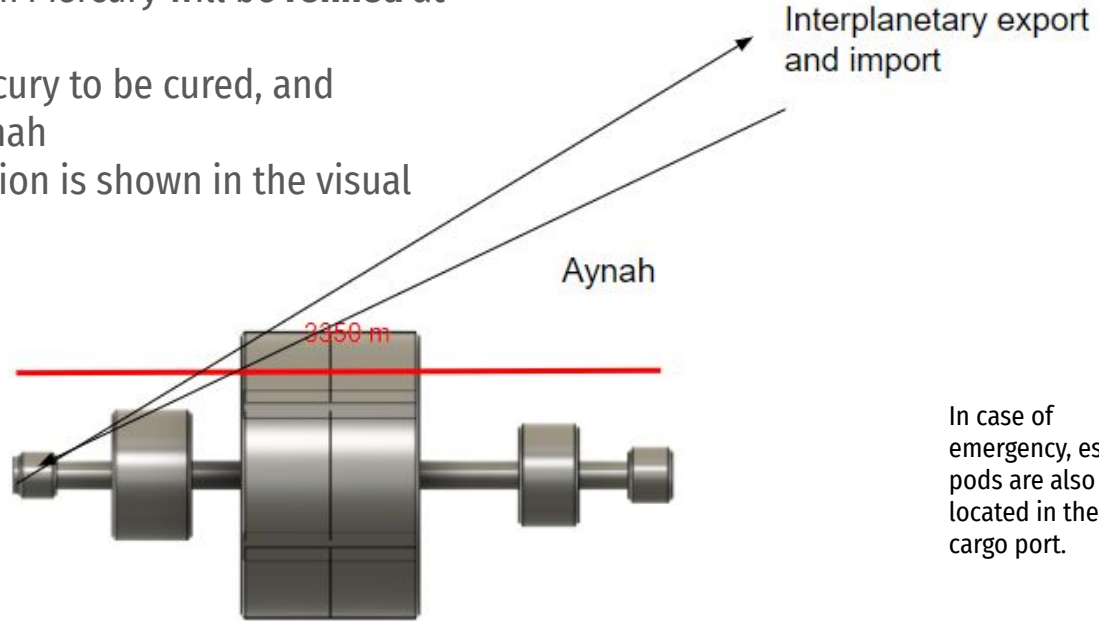
Diagrams by Amanda Nguyen

Aynah and Interplanetary Transport

- Materials (reardonium) from Mercury will be **refined** at the Cargo Port on Aynah
- Then, it will be sent to Mercury to be cured, and eventually sent back to Aynah
- The network of transportation is shown in the visual



Sun



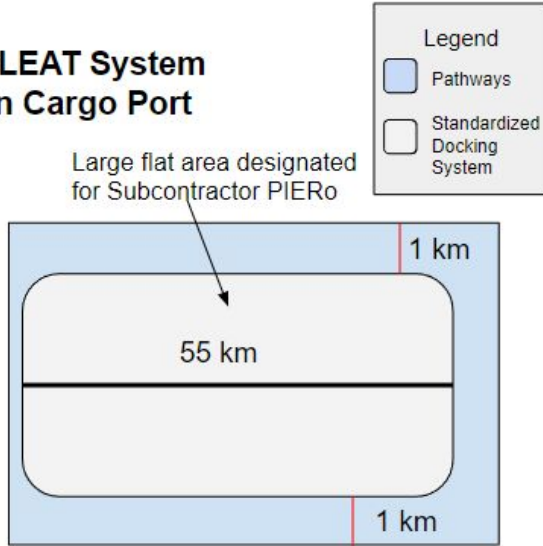
In case of emergency, escape pods are also located in the cargo port.

**Figures not drawn to scale

Diagrams by Amanda Nguyen

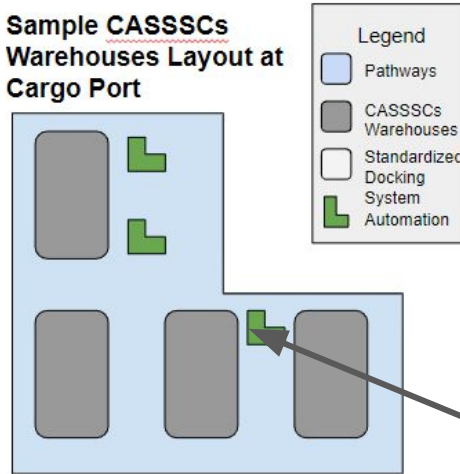
CASSSCs Warehousing and Automation at Cargo Port

CLEAT System on Cargo Port

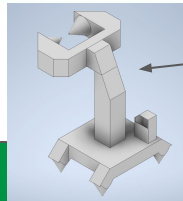


CLEAT system: large surface to accommodate the requirement for setting up the standardized docking system from new subcontractor PIERo.

Sample CASSSCs Warehouses Layout at Cargo Port



Automation: Beacham & d'Anconia

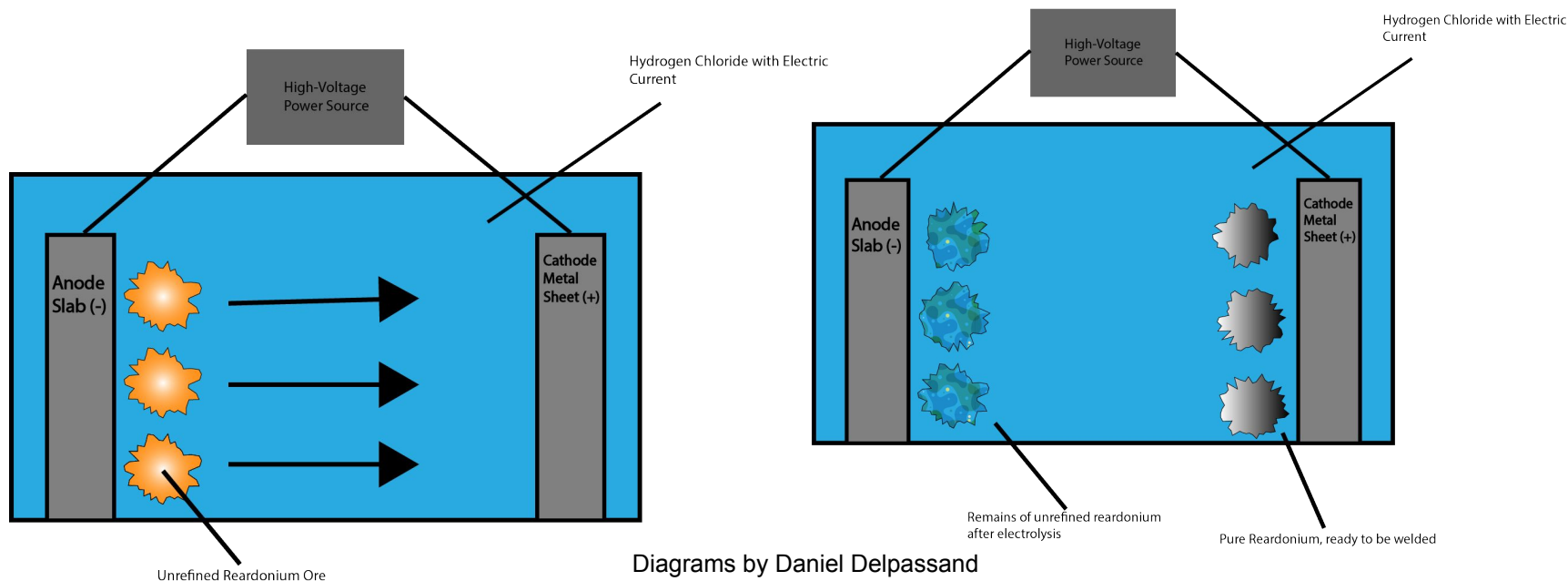


- The Automation shown operating at the port is **Beacham** mentioned earlier with the purpose of transport of material
- The diagram to the side is a visual for a sample layout at the Cargo Port
- It contains CASSSCs **warehouses** and the **standardized docking system flat area** for subcontractor PIERo

Diagrams by Amanda Nguyen

Infrastructure for refining and manufacturing reardonium parts

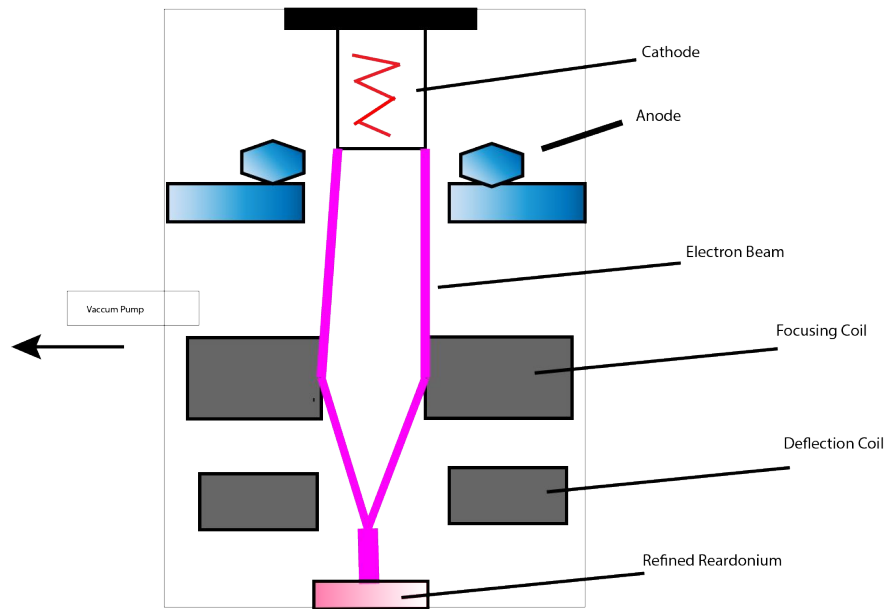
- In 0.5 G manufacturing area, electrolysis will be used to refine reardonium ore, using anode slab and cathode metal sheet



Diagrams by Daniel Delpassand

Infrastructure for refining and manufacturing reardonium parts (Cont)

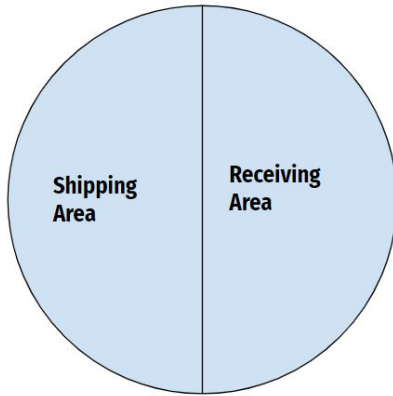
- In the 0 G manufacturing ring: Electron-beam welding molds refined reardonium into uniform, transportable units
- d'Anconia automations then load these transportable units into CASSSCs and onto shuttles.



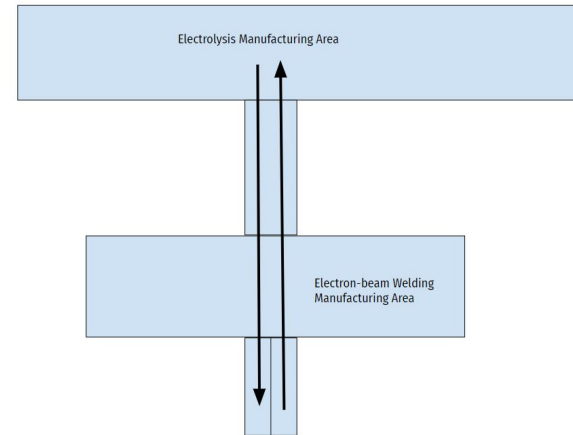
Diagrams by Daniel Delpassand

Receiving and Shipping

- Port closest to manufacturing ring split into 2 halves, one for receiving and one for shipping
- Within the receiving area, d'Anconia units sort CASSSCs into cured parts and raw ore during unloading
- McNamara units check the quality of both ore and cured parts
- Airlock in central axis between manufacturing and main rings prevents dust and grit from entering inhabitants' areas.



Diagrams by Daniel Delpassand



Gantt Chart

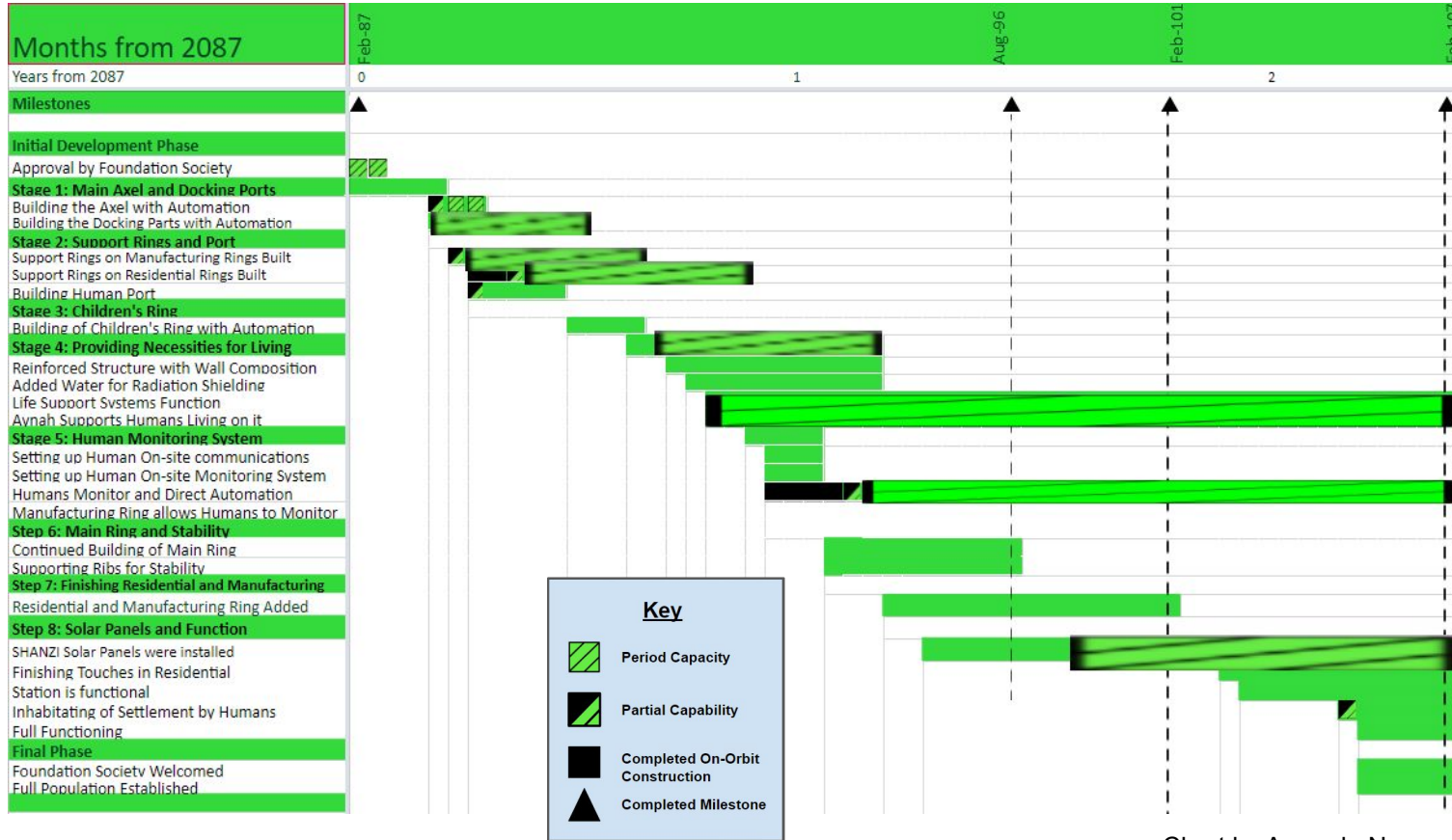


Chart by Amanda Nguyen

Costs



Structures

Tantalum Carbide	\$235,750,000,000
Titanium	\$47,146,666,000
Buckystructure	\$5,633,600,000
Water	\$1,450,000
Labor/Work	\$200,000,000
Total	\$288,000,000,000

Human Factors

Home Units	\$5,000,000,000
Recreational Centers	\$12,000,000,000
Space suits	\$3,000,000,000
Total	\$20,000,000,000

Automation

Auto Bots	\$163,004,000
Wristwatch	\$135,000
Mobile Tablet	\$540,000
Personal Laptop	\$1,120,000
VR Set	\$1,960,000
Total	\$166,759,000

Operations

Food	\$1080000000
Air	\$38,256,260,000
ZAP Factories	\$80,000,000
Yearly Energy Production	\$34,150,860
Total	\$39,450,410,860

Structures	\$288,000,000,000
Human Factors	\$20,000,000,000
Automation	\$166,759,000
Operations and Infrastructure	\$39,450,410,860
<u>Total</u>	<u>\$3,476,171,699,000</u>

Cost Tables by Daniel Delpassand